Learning Mixture of Unknown Causal Interventions

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Motivation (Causal Discovery)
Discovering Gene Regulatory Network

Problem:

1. **Interventions are noisy in real world experiments** (generally referred as fat-hand effect, imperfect or stochastic intervention)
2. **Noise in intervention could itself be random → leading to mixtures**

(unknown to us)

Intervention Experiment 1
Intended location : gene 1
Effective (mouse 1) : gene 1, gene 5
Effective (mouse 2) : gene 1, gene 4

Creates a mixture of unknown interventions

Intervention Experiment 2
Intended location : gene 3
Effective (mouse 5): gene 4, gene 1
Effective (mouse 3): gene3, gene 2

Goal : To find the gene-regulatory network for a given mouse species (underlying causal relationship)

Fundamental Problem: Observational Data are (generally) not sufficient to identify the causal graph.

Prior Work

Goal

Mixture Of Intervention → 1 ? → Disentangle Mixture (identify Intervention Targets and distributions) → 2 ✓ → Identify the Causal Graph

Disentangling Mixture of Intervention
(Kumar et. al., UAI-2021)

Contribution:
1. **Identification:** Recovered the unknown intervention targets
2. **Disentanglement:** Probability of each sample belonging to different interventional distribution.

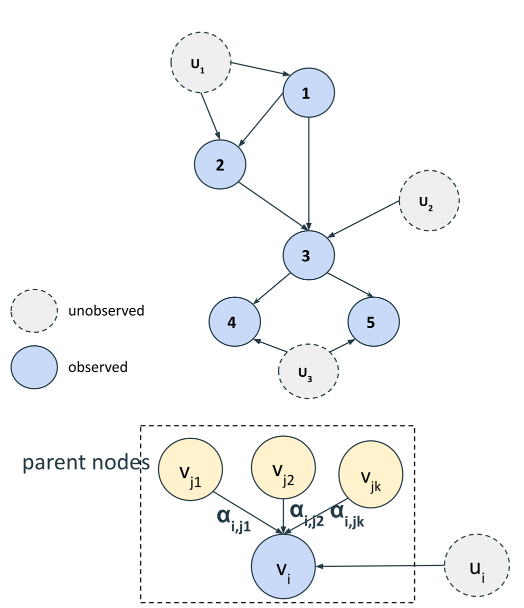
Assumptions:
1. Topological Order is given ⇒ cannot perform Step 2
2. No unknown confounders
3. The variables are discrete

Permutation-based causal structure learning with unknown intervention targets
(Squires et. al. UAI-2020)

Contribution:
1. Identify the unknown intervention targets (consistency under under mild assumption)
2. Identify the causal graph (equivalence class same as I-MEC)

Assumptions:
1. Mixture is already disentangled (requires Step 1)

Background



Structural Equation Model:
1. Defined by 3-tuple **M** = <V,U,F>
2. V = $\{V_1, V_2, \ldots, V_n\}$: set of observed nodes
3. U = $\{U_1, U_2, \ldots, U_m\}$: set of unobserved nodes
4. F = $\{f_1, f_2, \ldots, f_n\}$: set of deterministic equation describing causal relationship between random variables.

$$v_i = f_i(\boldsymbol{pa}(V_i), \boldsymbol{u}_i),$$

Linear SEM (with causal sufficiency): Identifying causal graph requires identifying A
1. f_i : linear function of parents
2. No-hidden confounder

$$v_i = f_i(\boldsymbol{pa}(V_i), \boldsymbol{u}_i) = \sum_{v_j \in \boldsymbol{pa}(V_i)} \alpha_{ij} v_j + u_i,$$

$$\boldsymbol{v} = A\boldsymbol{v} + \boldsymbol{u} \implies \boldsymbol{v} = (I - A)^{-1}\boldsymbol{u},$$

Linear SEM with Gaussian Additive noise

1. Gaussian noise: $u_i \sim N(\mu_i, \sigma_i)$
2. P(**u**) = N(**μ**,D) : noise distribution
3. Observational Distribution: P(**v**) = N(**m**,S)
   a. **m**=$(I-A)^{-1}$**μ** = B**μ**
   b. **S** = BDB^T

Casual Models governed by Linear SEM with Additive Gaussian noise are not identifiable using only the observational data.

Interventional Distribution

$$v_i = \sum_{v_j \in \boldsymbol{pa}(V_i)} \alpha_{ij}' v_j + u_i'$$

where $u_i' \sim N(\mu_i', \sigma_i')$ and α_{ij} can be changed

Type of Interventions:
1. **shift**: add a constant ⇒ $\mu_i' = \mu_i + \kappa$
2. **stochastic do**: $\alpha_{ij}' = 0$ and $u_i' \sim N(\mu_i', \sigma_i')$
3. **do:** $\alpha_{ij}' = 0$ and $u_i' \sim N(\mu_i', 0')$

Atomic Intervention:
Only one node is intervened at a time

Problem Formulation: Mixture Of Soft Atomic Intervention

$$\mathbb{P}_{mix}(\boldsymbol{V}) = \sum_{i_k \in \mathcal{I}} \pi_{i_k} \mathbb{P}_{i_k}(\boldsymbol{V})$$

Mixture Distribution (sample access)

Intervention Targets (unknown)

Mixing Coefficient (unknown)

$$\boldsymbol{u}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, D_i)$$
$$\boldsymbol{\mu}_i = \boldsymbol{\mu} + \gamma_i \boldsymbol{e}_i$$
$$D_i = D - \delta_i \boldsymbol{e}_i \boldsymbol{e}_i^T \text{ where } \delta_i \triangleq (\sigma_i - \sigma_i')$$

$\mathbb{P}_i(\boldsymbol{V}) = \mathcal{N}(\boldsymbol{m}_i, S_i)$ where

$$\boldsymbol{m}_i \triangleq B_i \boldsymbol{\mu}_i,$$
$$S_i \triangleq B_i D_i B_i^T$$
$$B_i \triangleq (I - A + \boldsymbol{e}_i \boldsymbol{c}_i^T)^{-1}$$

Interventional Distribution (parameters unknown)

Questions:
1. Does there exist **an algorithm** that can uniquely identify the parameters of mixture under infinite sample limit?

$$\theta = \{(\mathbf{m}_{i_k}, S_{i_k}, \pi_{i_k})\}_{i_k \in \mathcal{I}}$$

2. What is the **runtime and sample complexity** of such an algorithm?

Main Result

1

Theorem 1 (Disentangling Mixture):

Assuming causal graph is faithful and atleast one of the **control parameters** of intervention is not zero. Then, there exist an efficient algorithm that:
1. runs in time polynomial in "n" (number of nodes)
2. requires #samples = $poly\Big(n, \frac{1}{\epsilon}, \frac{1}{\delta}, \frac{1}{\min\left(\{poly(|\mathbf{c}_{i_k}|, \delta_{i_k}, \gamma_{i_k}, \frac{1}{|A_F|})\}_{i_k \in \mathcal{I}}\right)}\Big)$

with probability >= (1-δ) s.t. the recovered mixture parameters:

$$\sum_{i_k \in \mathcal{I}} \left(\left|\mathbf{m}_{i_k} - \hat{\mathbf{m}}_{\rho(i_k)}\right|^2 + \left|S_{i_k} - \hat{S}_{\rho(i_k)}\right|^2 + \left|\pi_{i_k} - \hat{\pi}_{\rho(i_k)}\right|^2 \right) \le \epsilon^2$$

$|c_{i_k}|$ = change in edge strength; δ_{i_k} = change in the noise variance; γ_{i_k} = change in the noise mean

2

Corollary 1 (Mixture-MEC = I-MEC):

Given samples from mixture of interventions $\mathbf{P}_{mix}$(**V**) over Linear-SEM with additive Gaussian noise and samples from the observational distribution **P**(**V**) we can identify the I-MEC of the underlying causal graph.

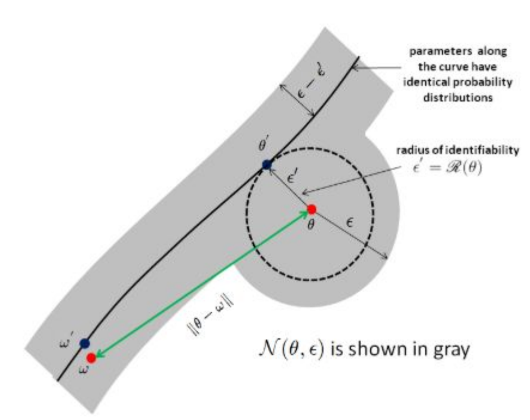
Control Parameters in different kind of intervention:
1. shift: $|\gamma_{i_k}| \ne 0$
2. stochastic do: $|c_{i_k}| \ne 0$
3. do: $|c_{i_k}| \ne 0, \delta_{i_k} \ne 0$

Proof Sketch

Prior Work (Mixture of Gaussian)

Radius of Identifiability (Belkin et al. 2010)

$$\mathcal{R}(\theta) = sup\,\{r > 0 \mid \forall \theta_1 \ne \theta_2, (|\theta_1 - \theta| < r, |\theta_2 - \theta| < r) \implies (\mathbb{P}_{\theta_1} \ne \mathbb{P}_{\theta_2})\}$$



Learning Mixture of Gaussian (Theorem 3.1 in Belkin et al. 2010)

There exist an efficient algorithm that:
1. runs in time polynomial in "n" (number of nodes)
2. requires #samples = $poly(n, \max(\frac{1}{\epsilon}, \frac{1}{\mathcal{R}(\Theta)}), \frac{1}{\delta}, Q)$

with probability >= (1-δ) s.t. the recovered mixture parameters:

$$\sum_{i_k \in \mathcal{I}} \left(\left|\mathbf{m}_{i_k} - \hat{\mathbf{m}}_{\rho(i_k)}\right|^2 + \left|S_{i_k} - \hat{S}_{\rho(i_k)}\right|^2 + \left|\pi_{i_k} - \hat{\pi}_{\rho(i_k)}\right|^2 \right) \le \epsilon^2$$

where, $(\mathcal{R}(\Theta))^2 \ge min\Big(\frac{1}{4} min_{i \ne j}(|\mathbf{m}_i - \mathbf{m}_j|^2 + |S_i - S_j^2|), min_i \pi_i\Big)$

Bounding Radius of Identifiability

Lemma 1: (Parameter Separation)

$$\|S_i - S_j\|_F^2 + \|\boldsymbol{m}_i - \boldsymbol{m}_j\|_F^2 \ge f(B, D)\Big(\|\boldsymbol{c}_i\|^2 + \|\boldsymbol{c}_j\|^2\Big) + h(B, D, \boldsymbol{\mu})\Big(\gamma_i^2 + \gamma_j^2\Big) + g(B)\Big(|\delta_i| \min\big(|\delta_i|, \lambda_{min}(D)\big) + |\delta_j| \min\big(|\delta_j|, \lambda_{min}(D)\big)\Big)$$

Intervention Parameters (we can control)

System Parameters (predefined)

Separation is poly function of the intervention parameter

Lemma 2: (Covariance Separation)

$$\|S_i - S_j\|_F^2 \ge \frac{\lambda_{min}^2(D)\Big(\|\boldsymbol{c}_i\|^2 + \|\boldsymbol{c}_j\|^2\Big)}{4\|B^{-1}\|_F^4} + \frac{|\delta_i|}{4\|B^{-1}\|_F^4} \min\Big(|\delta_i|, \lambda_{min}(D)\Big) + \frac{|\delta_j|}{4\|B^{-1}\|_F^4} \min\Big(|\delta_j|, \lambda_{min}(D)\Big),$$

Denser the graph lower the separation

Lemma 3: (Mean Separation)

$$\|\boldsymbol{m}_i - \boldsymbol{m}_j\|_F^2 \ge \begin{cases} \frac{\gamma_i^2}{4\|B^{-1}\|_F^2} + \frac{\gamma_j^2}{4\|B^{-1}\|_F^2}, & \psi_i^+, \psi_j^+ \text{ are active} \\ \frac{\gamma_i^2}{4\|B^{-1}\|_F^2}, & \psi_i^+, \psi_j^- \text{ are active and } \frac{\gamma_j^2}{4\|B\boldsymbol{\mu}\|^2} \le \|\boldsymbol{c}_j\|^2 \\ \frac{\gamma_j^2}{4\|B^{-1}\|_F^2}, & \psi_i^-, \psi_j^+ \text{ are active and } \frac{\gamma_i^2}{4\|B\boldsymbol{\mu}\|^2} \le \|\boldsymbol{c}_i\|^2 \\ 0, & \psi_i^-, \psi_j^- \text{ are active and } \frac{\gamma_i^2}{4\|B\boldsymbol{\mu}\|^2} \le \|\boldsymbol{c}_i\|^2, \frac{\gamma_j^2}{4\|B\boldsymbol{\mu}\|^2} \le \|\boldsymbol{c}_j\|^2 \end{cases}$$

where $\psi_i^+ \triangleq \Big(|\boldsymbol{c}_i^T B\boldsymbol{\mu}| < \frac{|\gamma_i|}{2} \text{ or } |\boldsymbol{c}_i^T B\boldsymbol{\mu}| > \frac{3|\gamma_i|}{2}\Big)$, $\psi_i^- \triangleq \frac{|\gamma_i|}{2} \le |\boldsymbol{c}_i^T B\boldsymbol{\mu}| \le \frac{3|\gamma_i|}{2}$ *and similarly for* ψ_j^+ *and* ψ_j^-. *If*

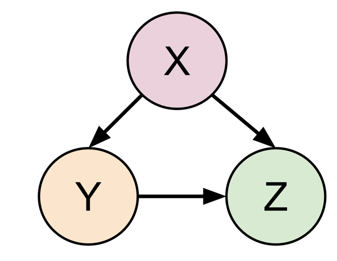
When the mean perturbation (γ) is small the mean separation ($|\mathbf{m}_i - \mathbf{m}_j|$) could be zero **BUT**
In those cases the strength perturbation (**c**) is large

Counterexample: Mixture is not always identifiable

$$X = \mathcal{N}(0, 1)$$
$$Y = X$$
$$Z = X - Y + \mathcal{N}(0, 1)$$

(violates faithfulness)

Let mixture contains two component:
1. observational
2. stochastic do intervention on node Z
   a. The mean and variance of noise distribution is kept same i.e N(0,1)



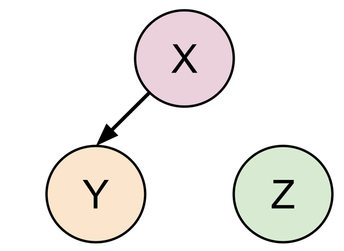
Observational Distribution

Both component have same mean and covariance matrix thus **zero parameter separation** in spite of the adjacency matrix being different.

$$|c_Z| \ne 0, |\gamma_{X/Y/Z}| = 0, \delta_{X/Y/Z} = 0$$

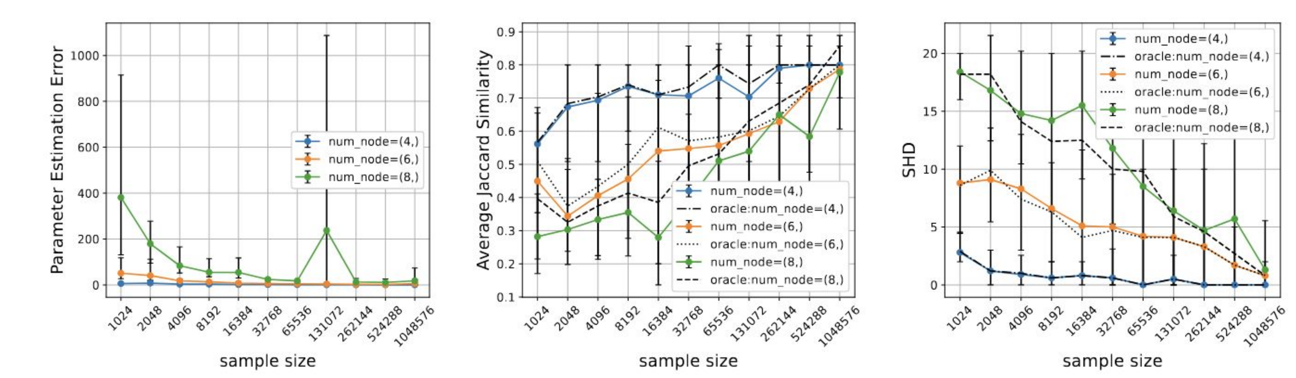
Our parameter separation captures such nuances explicitly.

$$f(B, D) = \lambda_{min}(D) = 0$$



Intervention on node Z

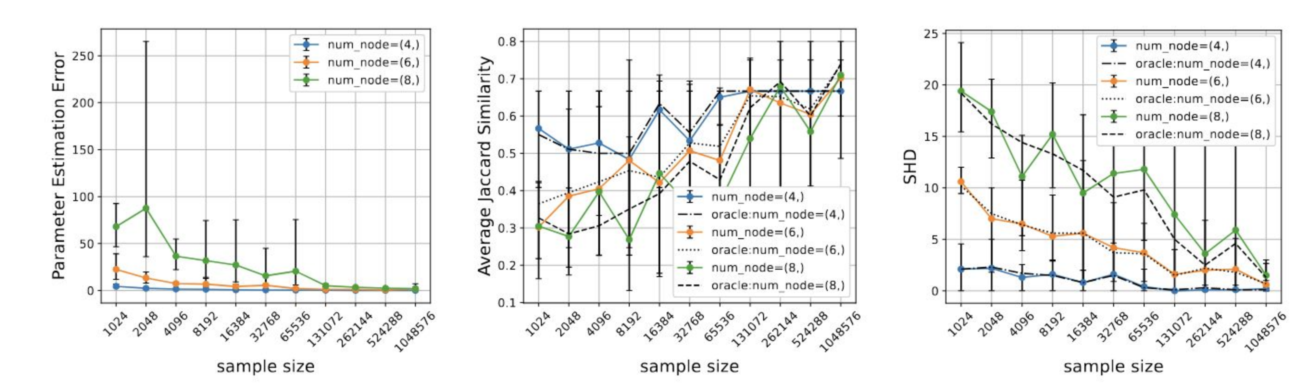
Simulation Study



(a) num intervention = all (↓) (b) num intervention = all (↑) (c) num intervention = all (↓)

Setting: All atomic interventional distribution are present in the mixture.

Observations: As sample size increases:
A. Mixture Parameter Estimation Error decrease → Mixture is disentangled
B. High Jaccard Similarity → Correct intervention targets are identified
C. Lower SHD → The estimated causal graph becomes close to ground truth

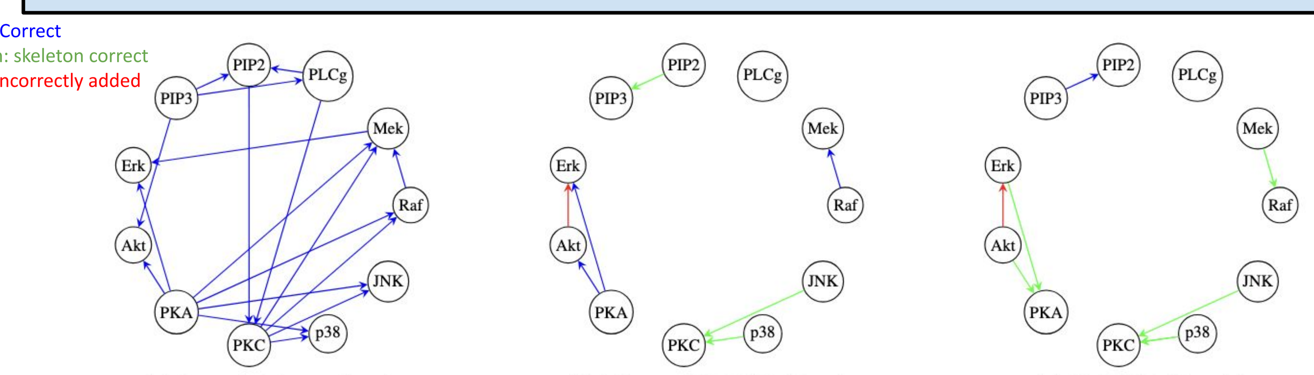


(d) num intervention = half (↓) (e) num intervention = half (↑) (f) num intervention = half (↓)

Setting: Half of all possible atomic interventional distribution are present in the mixture (selected randomly)

Semi-Synthetic Dataset
(Protein Signalling Dataset)

Blue: Correct
Green: skeleton correct
Red: incorrectly added



(a) Domain Expert Graph (b) Mixture-UTIGSP (Ours) (c) UTIGSP (Oracle)

cutoff ratio	Estimated/True Component	JS	Oracle JS	SHD	Oracle SHD
0.01	4/6	0.07	0.05	15	19
0.07	2/6	0.04	0.05	18	17
0.15	1/6	0.00	0.05	16	18
0.30	1/6	0.00	0.05	18	17
avg	2/6	0.03	0.05	16.75	17.75

K. Sachs, O. Perez, D. Pe'er, D. A. Lauffenburger and G. P. Nolan. Causal protein-signaling networks derived from multiparameter single-cell data. Science 308.5721 (2005): 523-529